HCIA-AI V4.0 Certification Practice Exam • Practice Set 1

Module 8: Practice Exam Set 1 (100 High-Yield Exam-Style Questions)

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Question 1

TRUE OR FALSE • 16 Points

In MindSpore, when constructing a custom neural network, developers define forward computation inside the backward() method and must not override construct().

A. True

B. False

Question 2

TRUE OR FALSE • 16 Points

In a neural network based on Connectionism, each individual neuron node strictly expresses a specific, isolated human semantic meaning.

A. True

B. False

Question 3

TRUE OR FALSE • 16 Points

Voice recognition (Automatic Speech Recognition) refers to the computational process of converting acoustic voice audio data into text data.

A. True

B. False

Question 4

TRUE OR FALSE • 16 Points

Huawei's Ascend 310 processor is designed only for massive cloud-scale training clusters, whereas Ascend 910 is an ultra-low-power chip designed for IoT edge watches.

A. True

B. False

Question 5

TRUE OR FALSE • 16 Points

'Batch inference is an offline batch job that performs model inference on bulk datasets. Therefore, there is no need for model training before executing batch inference.' This statement is correct.

A. True

B. False

Page Answers: **Q1**: False • **Q2**: False • **Q3**: True • **Q4**: False • **Q5**: False

Question 6

TRUE OR FALSE • 16 Points

In PyTorch, tensor calculations and neural network models are executed on the GPU by default without calling the .to(device) method.

A. True
B. False

Question 7

TRUE OR FALSE • 16 Points

In Python, tuples are mutable list-like objects whose elements can be freely appended, removed, and reassigned in-place after creation.

A. True
B. False

Question 8

TRUE OR FALSE • 16 Points

In Huawei's full-stack AI strategy, MindSpore is designed to support device, edge, and cloud collaborative AI development.

A. True
B. False

Question 9

TRUE OR FALSE • 16 Points

Overfitting can occur in both classification and regression machine learning problems.

A. True
B. False

Question 10

TRUE OR FALSE • 16 Points

In Huawei's Atlas computing platform, Atlas 500 is an AI cluster of supercomputers designed for deep underground nuclear simulation.

A. True
B. False

Page Answers: **Q6**: False • **Q7**: False • **Q8**: True • **Q9**: True • **Q10**: False

Question 11

TRUE OR FALSE • 16 Points

In MindSpore, the PyNative execution mode executes operations line by line like standard Python, providing convenient interactive debugging.

A. True
B. False

Question 12

TRUE OR FALSE • 16 Points

In NumPy, broadcasting allows arithmetic operations between arrays of differing shapes as long as their trailing dimensions are compatible.

A. True
B. False

Question 13

TRUE OR FALSE • 16 Points

In deep learning optimization, increasing the learning rate to an extremely high value guarantees that the model will always converge faster to the minimum loss.

A. True
B. False

Question 14

TRUE OR FALSE • 16 Points

In Huawei CANN architecture, TBE (Tensor Boost Engine) allows developers to write custom high-performance neural network operators for Ascend NPUs.

A. True
B. False

Question 15

TRUE OR FALSE • 16 Points

Huawei Atlas 900 is an AI cluster consisting of thousands of Ascend 910 processors, designed for scientific research and massive deep learning supercomputing.

A. True
B. False

Page Answers: **Q11**: True • **Q12**: True • **Q13**: False • **Q14**: True • **Q15**: True

Question 16

TRUE OR FALSE • 16 Points

In Python dictionary structures, mutable objects like lists can be safely used as dictionary keys.

A. True

B. False

Question 17

TRUE OR FALSE • 16 Points

Huawei HiLens is an end-cloud collaborative AI development platform providing multi-modal development tools and device-cloud synergy for vision applications.

A. True

B. False

Question 18

TRUE OR FALSE • 16 Points

In modern foundation models, Reinforcement Learning from Human Feedback (RLHF) completely replaces all pre-training data requirements.

A. True

B. False

Question 19

TRUE OR FALSE • 16 Points

In gradient descent optimization, saddle points are critical points where the first derivative (gradient) is zero, but which are neither local minima nor local maxima.

A. True

B. False

Question 20

TRUE OR FALSE • 16 Points

In Python, the list.append() method modifies the list in place and returns a new list containing the appended element.

A. True

B. False

Page Answers: **Q16**: False • **Q17**: True • **Q18**: False • **Q19**: True • **Q20**: False

Question 21

TRUE OR FALSE • 16 Points

Huawei Cloud ModelArts includes automated machine learning (AutoML) capabilities that automatically perform data preprocessing, feature engineering, and model selection.

A. True
B. False

Question 22

TRUE OR FALSE • 16 Points

The Sigmoid activation function is symmetric around the origin (0, 0) and has zero-centered outputs.

A. True
B. False

Question 23

TRUE OR FALSE • 16 Points

In Python, lambda expressions can contain multiple complex statements, loops, and exception handling blocks.

A. True
B. False

Question 24

TRUE OR FALSE • 16 Points

In MindSpore, Automatic Parallelism can automatically search for optimal parallel strategies (data, operator, and pipeline parallelism) without manual tensor slicing.

A. True
B. False

Question 25

TRUE OR FALSE • 16 Points

In convolutional neural networks (CNNs), pooling layers are used to reduce feature map spatial dimensions and help prevent overfitting.

A. True
B. False

Page Answers: **Q21**: True • **Q22**: False • **Q23**: False • **Q24**: True • **Q25**: True

Question 26

SINGLE-ANSWER QUESTION • 16 Points

Which of the following layers of Huawei's full-stack AI solution provides chip enablement and computing operator libraries for Ascend processors?

- **A.** Compute Architecture for Neural Networks chip enablement layer
- **B.** ModelArts one-stop AI application development cloud platform
- **C.** MindSpore unified training and inference deep learning framework
- **D.** MindX SDK application enablement layer

Question 27

SINGLE-ANSWER QUESTION • 16 Points

Consider a scenario where a machine learning algorithm is trained to filter spam emails. What constitutes the 'Experience (E)' in Mitchell's ML definition?

- **A.** The static network IP address assigned to the corporate email gateway server
- **B.** The raw execution throughput latency measured by the server operating system
- **C.** The percentage of spam messages correctly flagged by the classification model
- **D.** The historical database archive of labeled spam and legitimate email messages

Question 28

SINGLE-ANSWER QUESTION • 16 Points

A computer uses labeled images to learn and determine which pictures contain apples and which contain pears. Which type of machine learning does this represent?

- **A.** Reinforcement learning
- **B.** Unsupervised learning
- **C.** Supervised learning
- **D.** Semi-supervised learning

Question 29

SINGLE-ANSWER QUESTION • 16 Points

In the traditional AI business development process, which phase immediately follows Data Engineering and Feature Processing?

- **A.** Model Training and Validation
- **B.** Requirement Clarification
- **C.** Hardware Silicon Fabrication
- **D.** System Decommissioning

Question 30

SINGLE-ANSWER QUESTION • 16 Points

In Huawei Cloud, which service is paired with ModelArts to easily deploy trained vision AI models directly to edge devices?

- **A.** Object Storage Service
- **B.** HiLens
- **C.** Cloud Eye
- **D.** Elastic Cloud Server

Page Answers: **Q26**: A • **Q27**: D • **Q28**: C • **Q29**: A • **Q30**: B

Question 31

SINGLE-ANSWER QUESTION • 16 Points

Which of the following Atlas series hardware products is an AI accelerator card used in data center servers for high-throughput inference and video decoding?

A. Atlas 500
B. Atlas 200
C. Atlas 300
D. Atlas 900

Question 32

SINGLE-ANSWER QUESTION • 16 Points

Which of the following Atlas series hardware products is an embedded AI accelerator module designed for smart cameras, drones, and robotics?

A. Atlas 300
B. Atlas 800
C. Atlas 900
D. Atlas 200

Question 33

SINGLE-ANSWER QUESTION • 16 Points

In Python, which function is used to determine the number of items in a sequence (such as a string, list, or dictionary)?

A. count()
B. length()
C. len()
D. size()

Question 34

SINGLE-ANSWER QUESTION • 16 Points

Which of the following statements about Logistic Regression is correct?

A. Logistic regression is an unsupervised clustering algorithm optimizing intra-cluster variance
B. Logistic regression cannot solve binary classification problems without multi-layer perceptrons
C. Logistic regression requires Mercer kernel functions to compute gradients during backpropagation
D. Logistic regression applies the Sigmoid function to map linear combinations to probabilities

Question 35

SINGLE-ANSWER QUESTION • 16 Points

In neural networks, what is the vector of partial derivatives of a scalar loss function with respect to all parameters called?

A. Learning Rate
B. Momentum Factor
C. Gradient
D. Batch Size

Page Answers: **Q31**: C • **Q32**: D • **Q33**: C • **Q34**: D • **Q35**: C

Question 36

SINGLE-ANSWER QUESTION • 16 Points

In deep learning activation functions, which function maps any real value into a range between 0 and 1, commonly used in binary classification output layers?

- **A.** ReLU
- **B.** Tanh
- **C.** Leaky ReLU
- **D.** Sigmoid

Question 37

SINGLE-ANSWER QUESTION • 16 Points

In Python NumPy, what attribute gives the dimensional tuple of an array?

- **A.** ndarray.size
- **B.** ndarray.shape
- **C.** ndarray.dtype
- **D.** ndarray.ndim

Question 38

SINGLE-ANSWER QUESTION • 16 Points

In Python, what is the output of the slice expression a = [10, 20, 30, 40, 50]; print(a[1:4])?

- **A.** [20, 30, 40, 50]
- **B.** [10, 30, 50]
- **C.** [20, 30, 40]
- **D.** [10, 20, 30]

Question 39

SINGLE-ANSWER QUESTION • 16 Points

Which Huawei tool in ModelArts provides device-edge-cloud joint application development with hardware affinity and operator optimization?

- **A.** MindX DL
- **B.** MindSpore
- **C.** ModelBox
- **D.** AscendCL

Question 40

SINGLE-ANSWER QUESTION • 16 Points

What is the primary role of the DaVinci architecture inside Huawei Ascend AI processors?

- **A.** Managing cloud virtual machine billing and multi-tenant user authentication
- **B.** Providing front-end web visualization dashboards for data annotation teams
- **C.** Compiling high-level Python scripts directly into x86 desktop executables
- **D.** Providing dedicated 3D Cube matrix computing engines tailored for deep neural network operations

Page Answers: **Q36**: D • **Q37**: B • **Q38**: C • **Q39**: C • **Q40**: D

Question 41

SINGLE-ANSWER QUESTION • 16 Points

In MindSpore, which execution mode compiles the computation into a static computational graph for maximized execution performance?

- **A.** PyNative Mode
- **B.** Eager Script Mode
- **C.** Dynamic Trace Mode
- **D.** Graph Mode

Question 42

SINGLE-ANSWER QUESTION • 16 Points

Which of the following algorithms is based on Bayes' Theorem and assumes that feature attributes are conditionally independent of each other?

- **A.** Support Vector Machine
- **B.** K-Means
- **C.** Linear Regression
- **D.** Naive Bayes

Question 43

SINGLE-ANSWER QUESTION • 16 Points

In Python, which statement is used to handle exceptions and prevent program crashes upon errors?

- **A.** def ... return
- **B.** if ... else
- **C.** for ... in
- **D.** try ... except

Question 44

SINGLE-ANSWER QUESTION • 16 Points

Which of the following loss functions is most appropriate for training a multi-class neural network classifier?

- **A.** Mean Absolute Percentage Error
- **B.** Binary Hinge Loss Support Classifier
- **C.** Categorical Cross-Entropy Loss Model
- **D.** Mean Squared Error Continuous Loss

Question 45

SINGLE-ANSWER QUESTION • 16 Points

In Python, what is the data type of the variable x after executing x = (1, 2, 3)?

- **A.** dict
- **B.** tuple
- **C.** set
- **D.** list

Page Answers: **Q41**: D • **Q42**: D • **Q43**: D • **Q44**: C • **Q45**: B

Question 46

SINGLE-ANSWER QUESTION • 16 Points

Which Huawei development suite provides the Ascend deep learning development kit, compiler, and low-level runtime drivers?

- **A.** CANN Toolkit
- **B.** MindSpore Hub
- **C.** ModelArts Studio
- **D.** HarmonyOS SDK

Question 47

SINGLE-ANSWER QUESTION • 16 Points

In decision trees, what does Information Gain measure?

- **A.** The total quantity of leaf nodes generated after pruning cross-validation partitions
- **B.** The reduction in entropy achieved by partitioning data according to a specific feature
- **C.** The physical file storage footprint occupied by the serialized model checkpoint on disk
- **D.** The wall-clock execution duration required by the CPU compiler to finish tree training

Question 48

SINGLE-ANSWER QUESTION • 16 Points

In Python, which keyword is used to define a user-defined function?

- **A.** function
- **B.** sub
- **C.** def
- **D.** func

Question 49

SINGLE-ANSWER QUESTION • 16 Points

Which of the following operations in CNN reduces the spatial dimensions of feature maps while retaining dominant features?

- **A.** Padding
- **B.** Pooling
- **C.** Stride
- **D.** Convolution

Question 50

SINGLE-ANSWER QUESTION • 16 Points

In Python, what is the output of bool([])?

- **A.** False
- **B.** True
- **C.** Error
- **D.** None

Page Answers: **Q46**: A • **Q47**: B • **Q48**: C • **Q49**: B • **Q50**: A

Question 51

SINGLE-ANSWER QUESTION • 16 Points

What is the primary difference between a parameter and a hyperparameter in machine learning?

- **A.** Hyperparameters are learned automatically; parameters are configured manually
- **B.** Parameters are only used in regression; hyperparameters are only used in classification
- **C.** There is no conceptual difference between them
- **D.** Parameters are learned automatically from data; hyperparameters are configured manually before training

Question 52

SINGLE-ANSWER QUESTION • 16 Points

Which of the following is an example of an unsupervised learning task?

- **A.** Grouping customer transaction behaviors into clusters without ground-truth labels
- **B.** Classifying radiology images as tumor or healthy based on expert diagnoses
- **C.** Transcribing speech recordings into text
- **D.** Predicting tomorrow's temperature from historical records

Question 53

SINGLE-ANSWER QUESTION • 16 Points

What is the purpose of Early Stopping in deep learning?

- **A.** Increasing the optimizer learning rate exponentially at every training epoch
- **B.** Pruning low-magnitude convolutional filters during post-training quantization
- **C.** Terminating training when validation loss stops improving, thereby preventing overfitting
- **D.** Doubling the number of hidden layers dynamically whenever training loss plateaus

Question 54

SINGLE-ANSWER QUESTION • 16 Points

In Python, which module is part of the standard library and used for parsing regular expressions?

- **A.** re
- **B.** regex_lib
- **C.** text_match
- **D.** string_parser

Question 55

SINGLE-ANSWER QUESTION • 16 Points

Which of the following techniques converts categorical labels into binary vectors where only one index is 1 and all others are 0?

- **A.** One-Hot Encoding
- **B.** Min-Max Normalization
- **C.** Z-Score Standardization
- **D.** PCA Projection

Page Answers: **Q51**: D • **Q52**: A • **Q53**: C • **Q54**: A • **Q55**: A

Question 56

SINGLE-ANSWER QUESTION • 16 Points

In deep learning, what does the term 'Batch Size' refer to?

- **A.** The cumulative file count of raw image files stored inside the uncompressed dataset directory
- **B.** The number of training samples processed in one forward and backward gradient computation pass
- **C.** The maximum storage capacity of physical server solid-state storage expressed in gigabytes
- **D.** The total number of convolutional feature extraction layers stacked inside the neural model

Question 57

SINGLE-ANSWER QUESTION • 16 Points

Which of the following describes the learning rate in gradient descent?

- **A.** A hyperparameter controlling the step size taken along the negative gradient to update weights
- **B.** The clock frequency of the processor execution cores measured in gigahertz during training
- **C.** The total quantity of cross-validation fold partitions allocated for model evaluation
- **D.** The cumulative duration taken by the cluster to finish training expressed in clock hours

Question 58

SINGLE-ANSWER QUESTION • 16 Points

In Python, which operator is used to check if two references point to the exact same object in memory?

- **A.** equals
- **B.** is
- **C.** in
- **D.** ==

Question 59

SINGLE-ANSWER QUESTION • 16 Points

What is the primary function of the Activation Function in an artificial neural network?

- **A.** Calculating objective cross-entropy loss gradients against ground truth label targets
- **B.** Re-initializing all internal weight parameters uniformly to zero at every training epoch
- **C.** Introducing non-linearity into the network, enabling it to learn complex non-linear mappings
- **D.** Rescaling input image pixel dimensions to match standardized square aspect ratio templates

Question 60

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements regarding Support Vector Machines (SVM) are correct? (Multiple Choice)

- **A.** The objective of SVM is to find a hyperplane that maximizes the geometric margin between classes
- **B.** Support vectors are the closest training sample points to the separating hyperplane
- **C.** SVM cannot be applied to binary classification problems under any configuration
- **D.** Kernel functions allow SVM to handle non-linearly separable datasets

Page Answers: **Q56**: B • **Q57**: A • **Q58**: B • **Q59**: C • **Q60**: ABD

Question 61

MULTIPLE-ANSWER QUESTION • 18 Points

In Python NumPy library, which of the following functions are commonly used to initialize new multidimensional arrays?

- **A.** np.zeros() (creating an array populated completely with zeros)
- **B.** np.ones() (creating an array populated completely with ones)
- **C.** np.arange() (creating an array with evenly spaced values within a given interval)
- **D.** np.linalg.inv() (computes the multiplicative inverse of a square matrix)

Question 62

MULTIPLE-ANSWER QUESTION • 18 Points

In the Python Pandas data science library, which core data structures represent one-dimensional and two-dimensional labeled data?

- **A.** Series
- **B.** DataFrame
- **C.** Tensor
- **D.** Tuple

Question 63

MULTIPLE-ANSWER QUESTION • 18 Points

In Support Vector Machine optimization, what are the effects of adjusting the regularization penalty parameter C?

- **A.** A large C value assigns a high penalty to misclassified samples, resulting in narrower margins
- **B.** A small C value leads to higher model variance and an increased risk of overfitting
- **C.** A small C value tolerates more training classification errors in exchange for a wider margin
- **D.** Parameter C controls the trade-off between margin maximization and classification error minimization

Question 64

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements correctly describe training iterations and batches in mini-batch SGD?

- **A.** An Epoch is completed after evaluating a single mini-batch of training samples
- **B.** A Batch represents a subset of training data processed together in a single forward-backward pass
- **C.** An Iteration requires evaluating the entire training dataset from beginning to end
- **D.** The number of iterations per epoch equals total dataset size divided by batch size

Question 65

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements regarding the Random Forest ensemble algorithm are TRUE?

- **A.** It constructs an ensemble of diverse decision trees trained on bootstrap samples of the training data
- **B.** At each node split, it selects the best split from a randomly chosen subset of features
- **C.** It trains decision trees sequentially, with each tree fitting residual errors of previous trees
- **D.** Individual decision trees must be optimized using backpropagation gradient descent

Page Answers: **Q61**: ABC • **Q62**: AB • **Q63**: ACD • **Q64**: BD • **Q65**: AB

Question 66

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following aspects belong to Huawei's Full-Stack AI Solution?

- **A.** ModelArts
- **B.** Compute Architecture for Neural Networks
- **C.** Ascend
- **D.** MindSpore

Question 67

MULTIPLE-ANSWER QUESTION • 18 Points

Huawei Cloud ModelArts provides ModelBox for device-edge-cloud joint development. Which of the following are its core optimization policies?

- **A.** Mandatory cloud-only execution
- **B.** Hardware affinity
- **C.** Automatic segmentation of operators
- **D.** Operator optimization

Question 68

MULTIPLE-ANSWER QUESTION • 18 Points

Under what circumstances should Machine Learning be considered rather than traditional rule-based programming?

- **A.** When problem logic is too complex to be hand-crafted into explicit hand-crafted rules
- **B.** When large volumes of representative historical data are available to learn patterns
- **C.** When the task involves purely fixed arithmetic formulas that never change over time
- **D.** When zero data exists and the solution must execute strictly on physical relays

Question 69

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are classic application scenarios of AI in modern intelligent systems?

- **A.** Smart Homes and Internet of Things (IoT)
- **B.** Autonomous Driving and Intelligent Vehicles
- **C.** AI for Science (AI4Science) such as Pangu-Weather
- **D.** Manual paper file sorting without digital computers

Question 70

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following hardware devices belong to Huawei's Atlas AI computing platform series?

- **A.** Atlas 800 AI training/inference server
- **B.** Atlas 500 AI edge station
- **C.** Atlas 200 AI accelerator module
- **D.** Atlas 300 AI accelerator card

Page Answers: **Q66**: ABCD • **Q67**: BCD • **Q68**: AB • **Q69**: ABC • **Q70**: ABCD

Question 71

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are primary architectural features of Huawei's MindSpore AI framework?

- **A.** Unified front-end API enabling seamless Device-Edge-Cloud development
- **B.** Auto-Parallelism capability simplifying distributed training of large models
- **C.** Native support for both PyNative and Graph execution modes
- **D.** Requiring third-party proprietary closed-source compilers for model execution

Question 72

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are built-in immutable data types in Python?

- **A.** Tuple
- **B.** String
- **C.** List
- **D.** Dictionary

Question 73

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are built-in mutable data structures in Python?

- **A.** List
- **B.** Dictionary
- **C.** Set
- **D.** Tuple

Question 74

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following methods are commonly used in machine learning to handle imbalanced datasets?

- **A.** Adjusting class loss weights
- **B.** Undersampling majority class samples
- **C.** Duplicating majority class samples to increase class skew
- **D.** Oversampling minority class samples

Question 75

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are popular Python libraries used in machine learning and data science workflows?

- **A.** Scikit-learn
- **B.** NumPy
- **C.** Pandas
- **D.** Matplotlib or Seaborn

Page Answers: **Q71**: ABC • **Q72**: AB • **Q73**: ABC • **Q74**: ABD • **Q75**: ABCD

Question 76

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about PCA (Principal Component Analysis) are correct?

- **A.** It requires categorical ground-truth target labels for supervised training
- **B.** It is an unsupervised linear dimensionality reduction technique
- **C.** Principal components maximize the variance of the projected data
- **D.** It expands feature dimensions to infinity via the kernel trick

Question 77

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are common regularization strategies in deep learning?

- **A.** Early Stopping
- **B.** Dropout
- **C.** L2 regularization
- **D.** L1 regularization

Question 78

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about Huawei Ascend AI processors are correct?

- **A.** Ascend processors rely on x86 scalar instruction sets without matrix units
- **B.** Built upon the proprietary DaVinci 3D Cube computing architecture
- **C.** Ascend 910 is primarily designed for high-throughput AI model training
- **D.** Ascend chips are restricted only to consumer mobile phone microcontrollers

Question 79

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following components are part of Huawei's CANN software architecture?

- **A.** TBE
- **B.** Graph Engine
- **C.** MindSpore Hub
- **D.** AscendCL

Question 80

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are primary advantages of Decision Trees over complex black-box neural networks?

- **A.** Explicit rule-based tree interpretability
- **B.** Native handling of mixed feature scales
- **C.** Complete immunity to unpruned overfitting
- **D.** Zero-parameter non-linear curve fitting

Page Answers: **Q76**: BC • **Q77**: ABCD • **Q78**: BC • **Q79**: ABD • **Q80**: AB

Question 81

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about K-Means clustering are true?

- **A.** The algorithm is sensitive to initial centroid selection and outliers
- **B.** It automatically discovers the optimal value of K without user configuration
- **C.** It partitions data into K non-overlapping clusters
- **D.** It iteratively updates cluster centroids by computing sample means

Question 82

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are common kernel functions used in Support Vector Machines (SVM)?

- **A.** Linear Kernel
- **B.** Sigmoid Kernel
- **C.** Radial Basis Function
- **D.** Polynomial Kernel

Question 83

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are primary stages in the Machine Learning development lifecycle?

- **A.** Data Collection, Cleansing, and Preprocessing
- **B.** Feature Engineering, Model Training, and Validation
- **C.** Model Evaluation, Hyperparameter Tuning, and Deployment Serving
- **D.** Dynamic silicon etching of host GPU instruction microcode

Question 84

MULTIPLE-ANSWER QUESTION • 18 Points

In deep learning Foundation Models, which of the following techniques are commonly used to adapt a large pre-trained model to specific downstream tasks?

- **A.** Parameter-Efficient Fine-Tuning (PEFT / LoRA)
- **B.** Manual hard-coded rule rewriting
- **C.** Supervised Fine-Tuning (SFT)
- **D.** Random weight re-initialization of all transformer layers

Question 85

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are common activation functions used in Deep Learning?

- **A.** ReLU
- **B.** Tanh
- **C.** Sigmoid
- **D.** Leaky ReLU

Page Answers: **Q81**: ACD • **Q82**: ABCD • **Q83**: ABC • **Q84**: AC • **Q85**: ABCD

Question 86

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about Ensemble Learning algorithms are true?

- **A.** Boosting algorithms train models sequentially to reduce bias
- **B.** Bagging algorithms build multiple models in parallel on bootstrap samples to reduce variance
- **C.** Ensemble models can only combine decision trees and prohibit neural networks
- **D.** Random Forest is an exemplary Bagging algorithm, whereas GBDT is a Boosting algorithm

Question 87

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following evaluation metrics are specifically designed for continuous regression rather than classification?

- **A.** Classification Accuracy Rate
- **B.** Precision and Recall Score
- **C.** Mean Squared Error
- **D.** Mean Absolute Error

Question 88

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are foundational architectural principles that allow CNNs to process images efficiently compared to Fully Connected Networks?

- **A.** Local receptive fields
- **B.** Weight sharing
- **C.** Full connection
- **D.** Recurrent feedback

Question 89

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are operations supported by the Python string data type?

- **A.** string.join()
- **B.** string.strip()
- **C.** string.replace()
- **D.** string.split()

Question 90

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are core components and services of Huawei's Cloud EI (Enterprise Intelligence) ecosystem?

- **A.** ModelArts one-stop AI development and lifecycle platform
- **B.** Graph Engine Service for large-scale graph analysis
- **C.** Image Recognition, OCR, and Natural Language Processing Cloud APIs
- **D.** Relational Database Schema Migration Service

Page Answers: **Q86**: ABD • **Q87**: CD • **Q88**: AB • **Q89**: ABCD • **Q90**: ABC

Question 91

MULTIPLE-ANSWER QUESTION • 18 Points

In Huawei MindSpore framework, which of the following statements regarding neural network definition and execution are TRUE?

- **A.** Custom neural network models and custom layers must inherit from mindspore.nn.Cell
- **B.** The forward computation logic of the network must be implemented inside the construct member function
- **C.** MindSpore requires manual symbolic derivation of all backward gradient formulas
- **D.** MindSpore supports both PyNative and Graph execution modes

Question 92

MULTIPLE-ANSWER QUESTION • 18 Points

In Huawei Ascend AI computing architecture, which of the following components constitute the CANN (Compute Architecture for Neural Networks) software stack?

- **A.** Ascend Computing Language
- **B.** Tensor Boost Engine operator development and compilation tool
- **C.** Fusion Engine for automated graph and operator fusion optimization
- **D.** Graph Engine for execution graph optimization, allocation, and stream management

Question 93

FILL-IN-THE-BLANK QUESTION • 16 Points

Huawei's AI processor series built on the DaVinci architecture is named ____.

Answer: _____

Question 94

FILL-IN-THE-BLANK QUESTION • 16 Points

In the Transformer Self-Attention mechanism, the dot product between query matrix Q and key matrix K requires computing the ____ of matrix K (denoted as K^T).

Answer: _____

Question 95

FILL-IN-THE-BLANK QUESTION • 16 Points

Huawei's computing platform that includes AI modules, cards, edge stations, and servers is named ____.

Answer: _____

Page Answers: **Q91**: ABD • **Q92**: ABCD • **Q93**: Ascend • **Q94**: transpose • **Q95**: Atlas

Question 96

FILL-IN-THE-BLANK QUESTION • 16 Points

In Huawei Cloud, the end-to-end one-stop AI development and lifecycle management platform is called ____.

Answer: _____

Question 97

FILL-IN-THE-BLANK QUESTION • 16 Points

The tool in ModelArts that supports device-edge-cloud joint development and execution is called Model____.

Answer: _____

Question 98

FILL-IN-THE-BLANK QUESTION • 16 Points

In Python, the data structure defined with curly braces {key: value} is called a ____.

Answer: _____

Question 99

FILL-IN-THE-BLANK QUESTION • 16 Points

The DaVinci architecture features a specialized 3D ____ computing engine optimized for tensor matrix multiplications.

Answer: _____

Question 100

FILL-IN-THE-BLANK QUESTION • 16 Points

Huawei's edge-cloud collaborative vision AI development platform is called Hi____.

Answer: _____

Page Answers: **Q96**: ModelArts • **Q97**: Box • **Q98**: dictionary (or dict) • **Q99**: Cube • **Q100**: Lens